

ParaBank Automation - Test Plan

Project Overview:

This project aims to automate the testing of the ParaBank banking application using Selenium WebDriver, Java, TestNG, Maven, Jenkins, and GitHub. The automation framework is developed using the Page Object Model (POM) design pattern.

Objective:

The objective of this project is to automate important banking functionalities and reduce manual testing effort.

Modules Covered:

- User Registration
- Login
- Open Account
- Account Overview
- Transfer Funds
- Bill Pay

Tools used:

- Java
- Selenium WebDriver
- TestNG
- Maven
- Jenkins
- Github

Test Approach:

The framework follows the Page Object Model (POM) design pattern. Test scripts are created for positive test scenarios and known defect scenarios are maintained separately.



Test Environment:

- Operating System: macOS
- Browser: Chrome, Firefox
- Application URL: <https://parabank.parasoft.com/parabank/index.htm>

Test Scenarios:

Registration:

- Register a new user with valid details.

Login:

- Login with valid credentials.

Open Account:

- Create a new savings account.

Account Overview:

- Verify account details are displayed.

Transfer Funds:

- Transfer funds between accounts.

Bill Pay:

- Make bill payments successfully.

TC ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC_001	Registration	Register new user	Enter valid details and click Register	User registered successfully	User registered successfully	Pass
TC_002	Login	Valid Login	Enter valid username/password	User logged in	User logged in	Pass
TC_003	Login	Invalid Login	Enter invalid credentials	Error message displayed	Error message displayed	Pass
TC_004	Login	Empty Login	Leave fields blank and click Login	Validation message displayed	Validation message displayed	Pass
TC_005	Open Account	Create Savings Account	Open new account	Account created	Account created	Pass
TC_006	Account Overview	View Account Details	Navigate to Accounts Overview	Account details displayed	Account details displayed	Pass
TC_007	Transfer Funds	Valid Transfer	Transfer funds between accounts	Transfer successful	Transfer successful	Pass
TC_008	Bill Pay	Valid Bill Payment	Pay bill with valid amount	Payment successful	Payment successful	Pass

Defects:

- Same account transfer is allowed.
- Zero amount transfer is allowed.
- Zero amount bill payment is allowed.
- Negative amount bill payment is allowed.

Jenkins Integration:

Jenkins is integrated with the automation framework to execute automated test runs. The Jenkins pipeline is configured to pull the latest code from the GitHub repository, build the project using Maven, execute the TestNG test suite, and generate test reports. The build is triggered manually using the "Build Now" option in Jenkins.

The screenshot displays the Jenkins web interface for a project named "Capstone Project". The top navigation bar shows the Jenkins logo, the project name, and the build number "#15". A green checkmark icon indicates a successful build. The main content area is divided into a left sidebar with navigation links (Status, Changes, Console Output, Edit Build Information, Delete build, Timings, Git Build Data, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, Previous Build) and a main panel. The main panel shows the build status "Success" and the build number "#15 (10 Jun 2026, 14:14:45)". Below this, there is a section for "Build Artifacts" listing various files and their sizes, each with a "view" link. The artifacts include: bullet_point.png (356 B), collapseall.gif (157 B), emailable-report.html (9.49 KIB), failed.png (977 B), index.html (43.42 KIB), jquery-3.6.0.min.js (87.40 KIB), navigator-bullet.png (352 B), passed.png (1019 B), skipped.png (967 B), TEST-TestSuite.xml (38.68 KIB), testng-reports.css (5.23 KIB), testng-reports.js (3.53 KIB), testng-reports1.css (6.88 KIB), testng-reports2.js (3.53 KIB), testng-results.xml (36.51 KIB), and TestSuite.txt (269 B). To the right of the artifacts, it says "Started 20 hr ago" and "Took 1 min 40 sec". Below the artifacts, there is a section for "Started by user Surya Prakash J" and a "This run spent:" section with a list of times: 13 ms waiting, 1 min 40 sec build duration, and 1 min 40 sec total from scheduled to completion. At the bottom, there is a "git" section showing the revision "57aeaf1c4e191d4c5220d02172560fdeb72234e" and the repository "https://github.com/suryaprakashgithubj/Capstone_Project.git".

Jenkins / Capstone Project / #15

Status Success #15 (10 Jun 2026, 14:14:45) Add description Keep this build forever

Changes

Console Output

Edit Build Information

Delete build '#15'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Build Artifacts

File	Size	View
bullet_point.png	356 B	view
collapseall.gif	157 B	view
emailable-report.html	9.49 KIB	view
failed.png	977 B	view
index.html	43.42 KIB	view
jquery-3.6.0.min.js	87.40 KIB	view
navigator-bullet.png	352 B	view
passed.png	1019 B	view
skipped.png	967 B	view
TEST-TestSuite.xml	38.68 KIB	view
testng-reports.css	5.23 KIB	view
testng-reports.js	3.53 KIB	view
testng-reports1.css	6.88 KIB	view
testng-reports2.js	3.53 KIB	view
testng-results.xml	36.51 KIB	view
TestSuite.txt	269 B	view

Started by user Surya Prakash J

This run spent:

- 13 ms waiting;
- 1 min 40 sec build duration;
- 1 min 40 sec total from scheduled to completion.

Revision: 57aeaf1c4e191d4c5220d02172560fdeb72234e

Repository: https://github.com/suryaprakashgithubj/Capstone_Project.git

- refs/remotes/origin/main

Conclusion:

The ParaBank Automation Framework successfully automates major banking functionalities. The framework supports automated execution, reporting, cross-browser testing, and Jenkins integration.

Github Repository:

https://github.com/suryaprakashgithubj/Capstone_Project.git